

Principles of communication systems

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Homework #07 Solutions

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Problem 1

$$n = \lceil \log_2 L \rceil$$

closest integer that
is greater than or
equal to $\log_2 L$

1) $L = 2, n = 1$

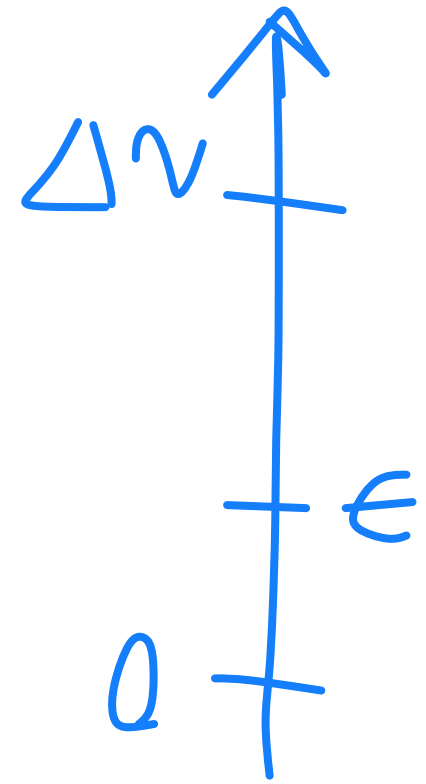
2) $L = 3, n = 2$

3) $L = 128, n = 7$

4) $L = 1023 = 2^{10} - 1, n = 10$

5) $L = 4097 = 2^{12} + 1, n = 13$

Problem 2



Mean square error : $\overline{\epsilon^2}$

$$\begin{aligned}\overline{\epsilon^2} &= \frac{1}{\Delta v} \int_0^{\Delta v} \epsilon^2 d\epsilon = \frac{1}{\Delta v} \cdot \frac{1}{3} \epsilon^3 \bigg|_0^{\Delta v} \\ &= \frac{1}{\Delta v} \cdot \frac{1}{3} \cdot \Delta v^3 = \frac{\Delta v^2}{3}\end{aligned}$$

Problem 3

$$1) \quad \text{SNR} = 10^2, \quad (\text{SNR})_{\text{dB}} = 10 \log_{10} 10^2 = 20 \text{ dB}$$

$$2) \quad \text{SNR} = \frac{1}{2}, \quad (\text{SNR})_{\text{dB}} = 10 \log_{10} 0.5 = -3 \text{ dB}$$

$$3) \quad \text{SNR} = 30, \quad (\text{SNR})_{\text{dB}} = 10 \log_{10} 30 = 14.8 \text{ dB}$$

$$4) \quad \text{SNR} = 1, \quad (\text{SNR})_{\text{dB}} = 10 \log_{10} 1 = 0 \text{ dB}$$

Problem 4

$$\gg \# \text{ of bits } n = \lceil \log_2 L \rceil = \lceil \log_2 2^7 \rceil = 7$$

$$\text{Signal bandwidth } B = 3.4 \text{ kHz}$$

$$\text{Sampling Rate } R = 2B = 6.8 \text{ kHz}$$

$$\# \text{ of bits/second} = \text{bit Rate } R_b = 2nB$$

$$\gg \text{Transmission bandwidth } B_T = \frac{R_b}{2} = nB = 23.8 \text{ kHz} \\ (\text{Max information channel})$$

$$\gg (SNR)_{dB} = 10 \log_{10} (C 2^{2n}) = 10 \log_{10} \left(\frac{3}{[\ln(1+\mu)]^2} \right) + 20n \log_{10} 2 = 33.6 \text{ dB}$$

Problem 5

Nyquist Rate $R_N = 2B = 14 \text{ kHz}$

Sampling Rate $R_S = \left(1 + \frac{25}{100}\right) R_N = 17.5 \text{ kHz}$

Max Quantization error $= \frac{\Delta v}{2} = \frac{m_P}{L} \leq 0.01 m_P \Rightarrow L \geq 100$

a) # of bits $n = \lceil \log_2 L \rceil = 7$

b) Total bit rate $R_b = n R_S = 7 \times 17.5 \text{ kHz} = 122.5 \frac{\text{kb}}{\text{s}}$